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Murtaza Pakawala

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Education

Sydney, Australia	UNSW	Feb 2022 - Dec 2025
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- Bachelor of Advanced Computer Science - First Class Honours
- Engineering WAM: 81
- Coursework: Machine Learning and Data Mining, Computer Vision, DeepLearning, Advance C++, Programming Challenges, Extended Operating Systems

Ann Arbor, USA	University Of Michigan	Jan 2025 - May 2025
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- Exchange Student - GPA: 3.7
- Coursework: Linear Algebra, Probability, Ethics in Technology

Research Thesis

UNSW Sydney	May 2025 – Dec 2025
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- **Thesis:** Inferring Evolutionary Parameters from Phylogenies.
- Developed a simulation-based inference pipeline using **msprime** and **PyTorch** to infer demographic parameters from phylogenetic trees under a four-parameter coalescent bottleneck model.
- Designed and compared multiple phylogeny representations, including summary-statistic and learned embedding approaches, to transform tree-structured data into fixed-length inputs for normalizing-flow-based posterior estimation.
- Evaluated models on large-scale simulated datasets across held-out parameter settings, analyzing posterior marginals and measuring absolute and relative error in both parameter space and summary-statistic space.

Personal Projects

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- **HalfCheetah Reinforcement Learning Project** [GitHub](#)
 - Implemented and analyzed reinforcement learning methods for the **HalfCheetah** continuous control environment, comparing approaches such as Monte Carlo, Temporal Difference, and Generalized Advantage Estimation within a policy optimization framework.
 - Explored how different return and advantage estimation strategies affect training stability, reward progression, and agent behavior, building a deeper practical understanding of modern reinforcement learning.
 - **C++ Neural Network from Scratch** [GitHub](#) Designed and implemented a neural network from scratch in C++, leveraging object-oriented programming (OOP) principles to modularize different components. Implemented gradient computation and backpropagation using the chain rule, providing a hands-on understanding of the inner workings of neural networks.

Extracurriculars

BLOGS

[Blog / Website](#)

- Write short blog posts on topics in mathematics and machine learning to explain concepts clearly and build intuition.

Skills

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- **Programming Languages:** C++, C, Java, Python, JavaScript, TypeScript, SQL
 - **Frameworks & Technologies:** React, React Native, Next.js, Express.js, NestJS, PostgreSQL

Awards

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- Second Prize in a personnel project competition organized by CSESOC
 - Highest Voted App prize organized by CSESOC